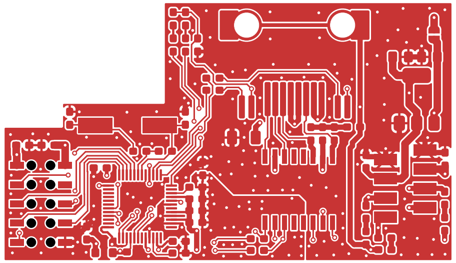
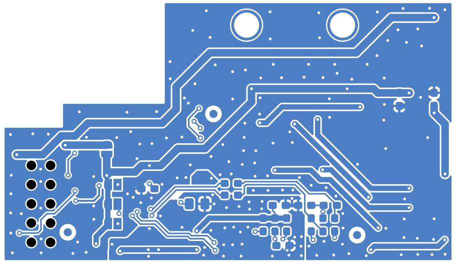
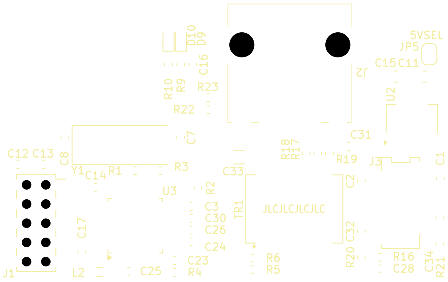
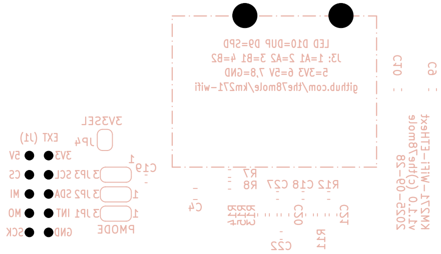
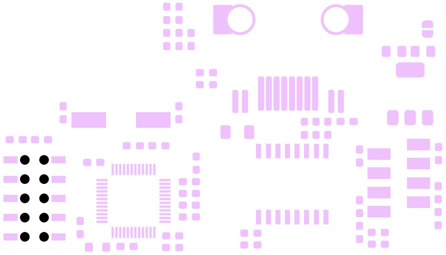


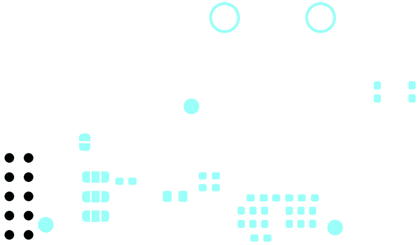


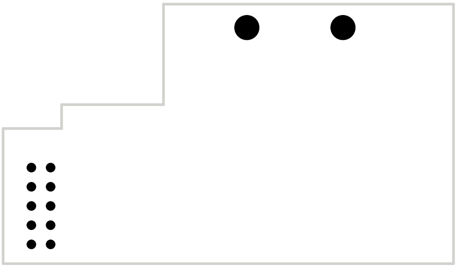


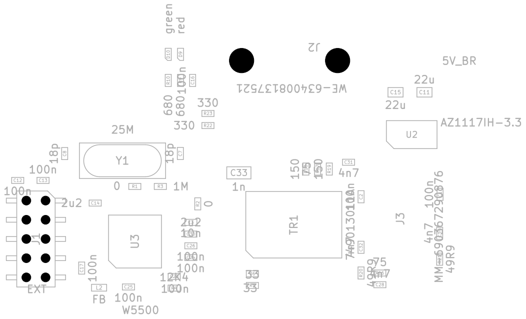
D10
D9
5VSEL
JP5
C16
C15
C11
J2
R10
R9
R23
R22
U2
C7
C31
C12
C13
C8
R18
R17
R19
J3
C1
Y1
C14
R1
R3
C33
R2
U3
C2
C3
C30
C26
C17
TR1
JLCJLCJLCJLC
C32
C24
R6
R16
C23
R20
C28
C34
R21
R5
J1
L2
C25
R4

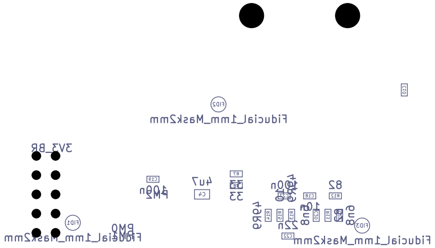

LED D10=DUP D9=SPD
J3: 1=A1 2=A2 3=B1 4=B2
5=3V3 6=5V 7,8=GND
github.com/the78mole/km271-wifi
C10
C9
KM271-WiFi-ETHext
v1.1.0 (c)the78mole
2025-09-28
3V3SEL
JP4
EXT (J1)
5V
3V3
CS
MI
MO
SCK
SCL
SDA
INT
GND
3 JP3
3 JP2
3 JP1
1
1
1
PMODE
C19
C4
R7
R8
C27
C18
R12
R13
R14
R15
C20
C21
C22
R11

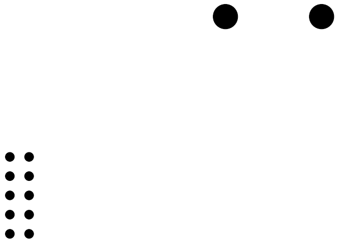







green
red
J2
5V_BR
WE-634008137521
22u
22u
680
680
100n
330
330
AZ1117IH-3.3
U2
25M
Y1
18p
18p
100n
100n
0
1M
C33
1n
150
75
150
4n7
2u2
0
TR1
J3
100n
4n7
U3
J1
2u2
10n
100n
100n
12k4
100n
100n
33
33
75
4n7
49R9
49R9
EXT
FB
100n
W5500


Fiducial_1mm_Mask2mm
3V3_BB
4u7
100n
PM2
PM0
82
6n8
22n
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm



FAB NOTES KM271–WiFi–ETHext rev v1.1.0
Stackup: 2 layers, FR4 TG135, 1.6 mm +–10 pct, outer copper 1 oz (35 um)
L1 F.Cu signal/GND pour | core 1.5 mm | L2 B.Cu signal/GND pour
Finish: HASL lead–free. Gold fingers: no
Mask green, silk white, vias tented. Min track/space 0.20/0.20 mm, min via 0.3/0.6 mm
Assembly: Standard PCBA, both sides. BOM/CPL: production/bom.csv, production/positions.csv